

ASANTE, Inc.- CU Boulder - Data Science



Capstone Project Description

Company Address: 6800 Wisconsin Ave. #1220, Chevy Chase, MD 20815.

Industry Description and Company Products:

ASANTE powers a community-centric ecosystem with a customizable rewards platform that builds connections between e-commerce and local communities. Please check our animation here: www.asan-te.com

The ASANTE **digital ecosystem** is designed to foster engagement and loyalty by bridging the gap between online retailers, their customer base and the local nonprofits they prefer to support. This infrastructure integrates advanced **data analytics and customizable loyalty rewards systems** to offer businesses, both large and small, tools to create personalized and cause-driven reward programs.

We are looking to work with students on designing/developing a data driven model that recommends nonprofits based on business preferences, client interests, and location; this model would facilitate nonprofit choice predictions for Businesses customizing their campaigns in the Campaign Manager setup flow. The goal is to prioritize nonprofit suggestions in the Search option in alignment with these factors, enhancing relevance and personalization over random selections.

Primary Contact(s)/Mentor(s) Names and Emails:

Matei Fawzy, CEO-CU Boulder Alum: mateifawzy@asanteinc.com

Dr. Camelia Fawzy, Internship program coordinator: camelia@asan-te.com

TJ Hershey, Product & Software Manager: tj.hershey@asan-te.com

Sawani Hejib, Intern-Database Developer: sawani.hejib@asan-te.com

Planned Meeting Schedule with Students

Meeting Venue: Zoom

Meeting Occurrence Rate: Weekly

Attention Measure: We are planning to work closely with the team(s) and provide access to project and communication channels for questions and documentation sharing as needed.

Requested Team Size: we believe a 5 team member team would be most effective for this project. We would also be excited to work with two-4 team members teams if there is a larger number of students interested in working on our project.

Specific Required/Expected Deliverables: none

Required Intellectual Property: none

Project Introduction

This project focuses on developing a data driven prediction solution that predicts the nonprofits most aligned with the business and customers based on business preferences, client interests, and location in the Campaign Manager setup flow. The goal is to predict and recommend the Campaign screen – nonprofit options those most aligned with these factors, enhancing relevance and personalization over random selections. The project involves several key steps:

Step 1. Web-Scraping: Web scraping techniques will be used to gather data from nonprofit and relevant product websites, including cause categories, locations, and impact reports. This data will be stored in a structured database for use in the AI recommendation model.

ASANTE will provide a list of businesses and product categories targeted for this process. *If a second team chooses this project, they will be provided with different lists.*

Data Cleaning & Preprocessing: The business client, and nonprofit data is cleaned and normalized to ensure consistency and accuracy, preparing it for analysis.

Step 2. Exploratory Data Analysis (EDA): EDA is performed to identify patterns and trends in business and client preferences, geographic proximity, and nonprofit engagement, providing deeper insights into the data. Create Dashboard for data visualization.

Step 3. Machine Learning modelling: Build and train a machine learning model using collaborative filtering, content-based filtering, and location-based ranking to generate nonprofit recommendations. By answering some research questions through visualizations such as: why is this model trained, what trends it represents, why this product is most purchased product, why do customers purchase this specific brand, what nonprofits are supported by customers purchasing which product, what nonprofits are supported by customers purchasing from which business.

Step 4. Search Engine Optimization: After cleaning the data, SEO can be integrated by optimizing content with relevant keywords, enhancing on-page elements like titles and meta descriptions, and using location data for local SEO. Developing high-quality, keyword-rich content about nonprofits and implementing schema markup helps improve search engine rankings and recommendations. Additionally, building backlinks and monitoring SEO performance ensures ongoing visibility and engagement. *If a second team chooses this project, both teams should have access to the combined data.*

Test the nonprofit AI suggestions in the campaign setup process.

List of Project Functionalities (Functional Elements – what will be created, built, examined, discovered, predicted, etc.)

Step 1: Web Scraping and Data Collection

Functionalities/Requirements:

Web Scraping:

Develop and implement web-scraping scripts to gather nonprofit and product data from target websites.

Extract data points such as cause categories, geographic locations, and impact reports from nonprofit sites.

Scrape relevant product data such as product categories, business preferences, and client interests from targeted business websites.

Data Storage:

Store all scraped data in a structured database to allow easy access for analysis.

Ensure data is formatted consistently and securely stored for further use.

Data Cleaning & Preprocessing:

Clean the raw scraped data to remove duplicates, missing values, and incorrect formats.

Normalize business, client, and nonprofit data, ensuring uniformity (e.g., consistent naming conventions for product categories, nonprofits, and locations).

Prepare the cleaned data for analysis and modeling by creating appropriate features and labels.

Step 2: Exploratory Data Analysis (EDA)

Functionalities/Requirements:

Data Exploration:

Conduct exploratory data analysis to understand the relationships between business preferences, client interests, nonprofit causes, and geographic proximity.

Identify patterns and trends in the data related to nonprofit engagement, customer behavior, and geographic influences.

Visualizations:

Create meaningful visualizations (e.g., bar charts, heat maps, geographic maps) to represent patterns and trends identified during EDA.

Use a visual tools like dashboards (e.g., Tableau, Power BI, or Plotly/Dash) to make insights accessible to non-technical users.

Dashboard Development:

Develop an interactive dashboard to display key insights, such as:

- Most supported nonprofit categories by business/product category.
- Popular products by customer preferences.
- Nonprofit categories engagement levels in different geographic areas.

Provide drill-down capabilities to explore the data by specific businesses, nonprofits, or locations.

Step 3: Machine Learning Modelling

Proposed Functionalities/Requirements:

Model Selection and Training:

Implement and train machine learning models, specifically:

Collaborative Filtering: Leverage user interaction data (e.g., business-client preferences) to suggest nonprofits.

Content-Based Filtering: Use features such as cause categories and geographic proximity to recommend nonprofits aligned with business/client preferences.

Location-Based Ranking: Rank nonprofits based on geographic proximity to the business or customers' preferences, factoring in regional relevance.

Proposed Model Evaluation:

Evaluate model performance using metrics such as precision, recall, F1 score, or ranking metrics (NDCG)-students will have flexibility in making their own decisions.

Visualize the model's performance and explain the reasoning behind its predictions through tools like confusion matrices, precision-recall curves, or SHAP values (for model interpretability).

Proposed Research Questions:

(we would like to discuss the questions with the students)

Address research questions with visualizations:

Why was this model trained?

What trends does the model represent?

Why is a specific product most purchased by customers with certain nonprofit preferences?

Which nonprofits are supported by customers purchasing which products?

What nonprofits are supported by customers purchasing from specific businesses?

Recommendation System Integration:

Integrate the trained recommendation system into the Campaign Manager setup process to provide personalized nonprofit recommendations to businesses based on data inputs.

Step 4: Search Engine Optimization (SEO)

Functionalities/Requirements:

Keyword Optimization:

Use insights from the data to identify relevant keywords for nonprofits and product categories. Optimize content with keywords, titles, meta descriptions, and headers on both nonprofit and product pages to enhance SEO.

Local SEO:

Incorporate geographic location data into the SEO strategy to improve search visibility for local nonprofit-business alignments.

Implement schema markup (e.g., for nonprofits, locations, causes) to ensure better search engine understanding of the content.

Testing AI Suggestions:

Test the integration of the AI-driven nonprofit recommendations within the Campaign Manager setup process, ensuring that suggested nonprofits are relevant and optimized for SEO.

List Dataset Needs and Availability

Students will be provided with some proprietary data, but they also need to gather their own during Step 1. Only publicly available data should be collected.

Project Notes, Requirements, Expectations

Step 1 Outcome: A fully populated database with clean, structured data that includes nonprofit and product/business information relevant to the model development process.

Step 2 Outcome: Insights and trends visualized on a dashboard that provides stakeholders with an understanding of customer preferences, business alignment, and geographic influences on nonprofit support.

Step 3 Outcome: A trained and evaluated machine learning model capable of recommending nonprofits based on client preferences, business profiles, and geographic proximity. Clear visualizations of model behavior and answers to key research questions.

Step 4 Outcome: Improved search engine visibility for the nonprofits and businesses through well-optimized content. The AI-based nonprofit recommendations tested and refined for relevance within the Campaign Manager setup process.

Project Background and Objective: Brief explanation of the company, project, and desired objective(s).

This project would support ASANTe engineers with making most informed decisions related to its future AI development. Our objective is to have students conduct research, complete the outcomes mentioned above and evaluate the process and outcomes at the end of the project.

Project Requirements: The project requirements and constraints should be stated in broad and simple terms. The specific details will be established between the client and the team.

There are no other project requirements besides the outcomes' expectations outlined above. We would like the team members to make their own decisions, explain why they choose certain tools and evaluate the outcomes in a final presentation; we would also like to have their recommendations for improvement if any.

Students working on this project will be required to sign an Internship Agreement with ASANTe (a draft will be provided for review to all students interested in joining a team).

Why students should choose this project:

We hope students will choose our ASANTe project for the following reasons:

1. **Real-World Application-** The project addresses a real business need and will help us develop an enhanced future matching process between businesses and customers with nonprofits that align with their values, preferences, and locations. This focus on corporate social responsibility (CSR) is very important for our loyalty rewards solution, and students can gain experience working on a project that delivers direct societal impact.
2. **Comprehensive Data Science Skill Set-** Our project covers the entire data science pipeline, providing a well-rounded learning experience.
3. **Personalization and Recommendation Systems-** Recommendation systems are widely used across industries (e.g., e-commerce, streaming platforms), and mastering them opens doors to numerous career opportunities. Students will gain practical experience in building a system that delivers personalized recommendations—an in-demand skill in the job market.
4. **High Impact and Visibility-** Our project can have a significant social impact by connecting businesses with nonprofits that align with their values and local communities. Students will contribute to a solution that could lead to better alignment between business actions and social good. If successful, the project will be implemented (entirely or partially) in one of our key business processes (Campaign Setup), giving students the satisfaction of seeing their work deployed in a live environment.
5. **Hands-On with Modern Tools and Techniques-** Students will have flexibility in choosing any cutting-edge data science tools and libraries such as Pandas, Scikit-Learn, SQL, Python Web Scraping Libraries (BeautifulSoup, Scrapy), visualization tools (e.g., Tableau, Power BI) and machine learning models they deem most appropriate for our project.

Deliverables: The client is asked to indicate the desired deliverables in the project. The final deliverables will be established through negotiation between the team and the client as a stage of the project.

Please consider the outcomes mentioned above as starting point for our deliverable's negotiation.

Skill Requirements: The client is asked to suggest the skill areas the students may need in order to complete the project.

Technical Skills: Web scraping, data cleaning, SQL/NoSQL, machine learning (collaborative/content-based filtering), data visualization and SEO.

Analytical Skills: EDA, data visualization, model evaluation, and dashboard creation.

Soft Skills: Teamwork, communication, project management, and critical thinking.

Images: Please include relevant images for this project. For example, these can be existing designs on which you would like the students to improve or photos of an existing apparatus with which they may work.

Other Questions to Consider and Note:

- This is an open-ended and discovery-based project with some development goals as described in the Outcomes.
- Students do not need to be US Citizens to work with us.
- We would like to work with students willing to learn and use AI tools to enhance performance on their project tasks when possible.
- We hope students are open to experiencing working in a fast paced, fully-remote environment with mentors on different time zones.
- We expect students to take responsibility for the tasks and inform team members of any delays that might impact the team ability to meet deadlines and deliver high quality outcomes.
- If students are open to taking a brief assessment, we would be happy to experiment with remote-team processes that might fit their individual work needs.